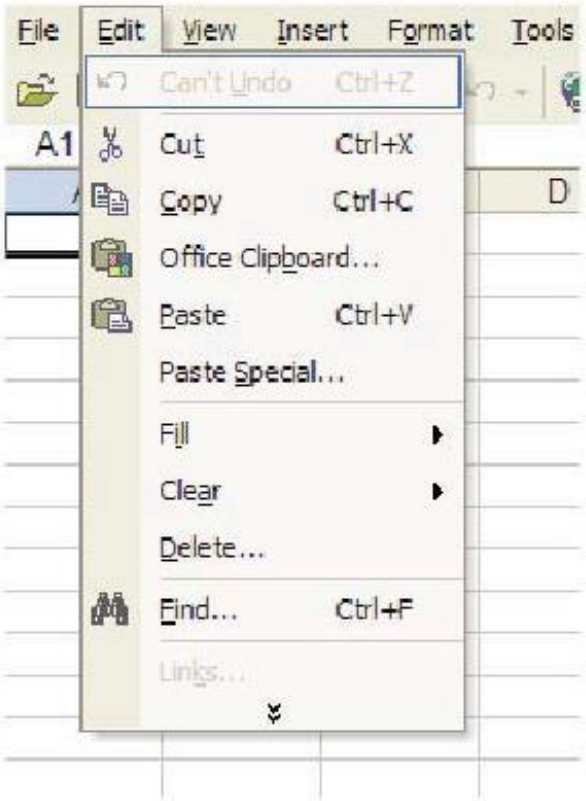
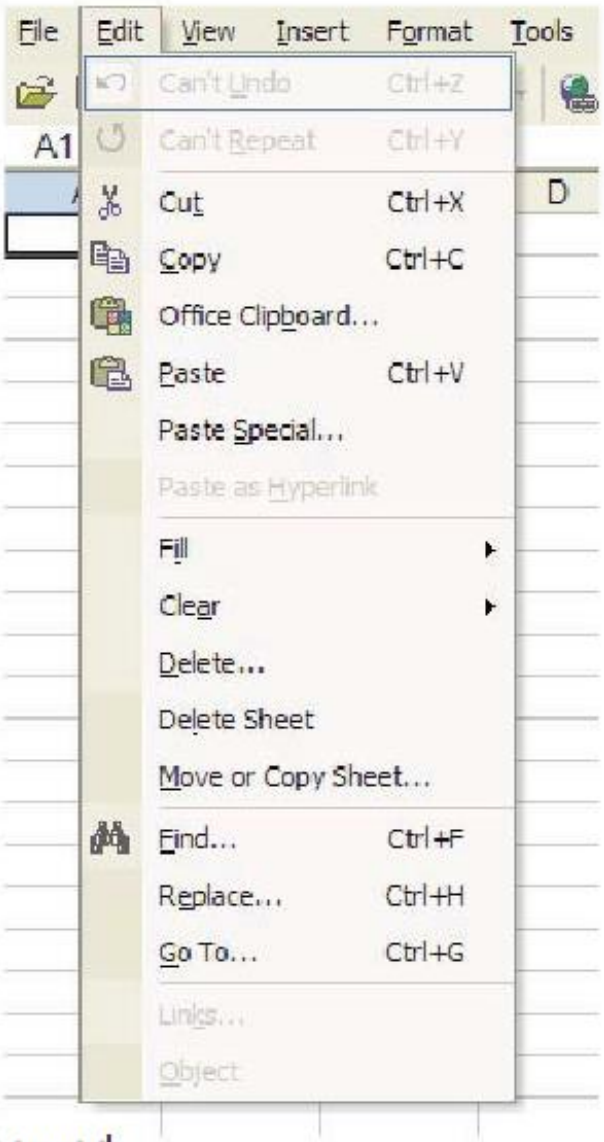
Controlled experiments

adaptive menus





Predictability: menu may change in complex and unpredictable ways – with new items appearing and underused items disappearing for no visible reason

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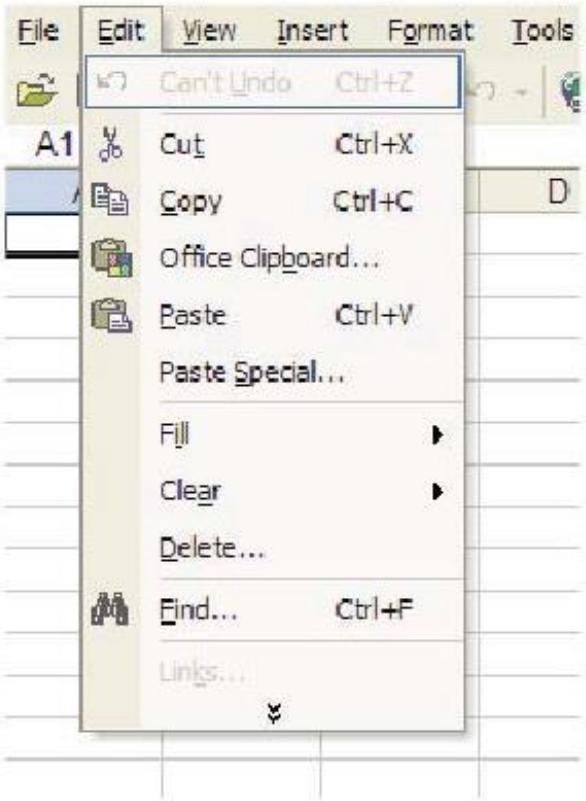
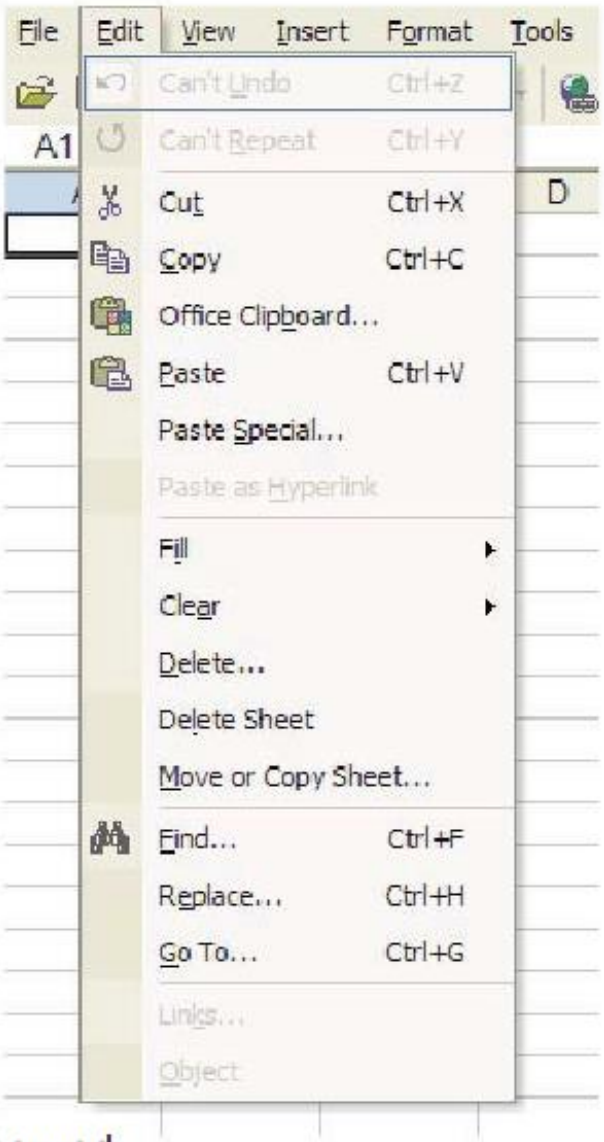
adaptive menus





user control: The adaptation happens automatically; the user can't manually fixate or remove items

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- Split menu

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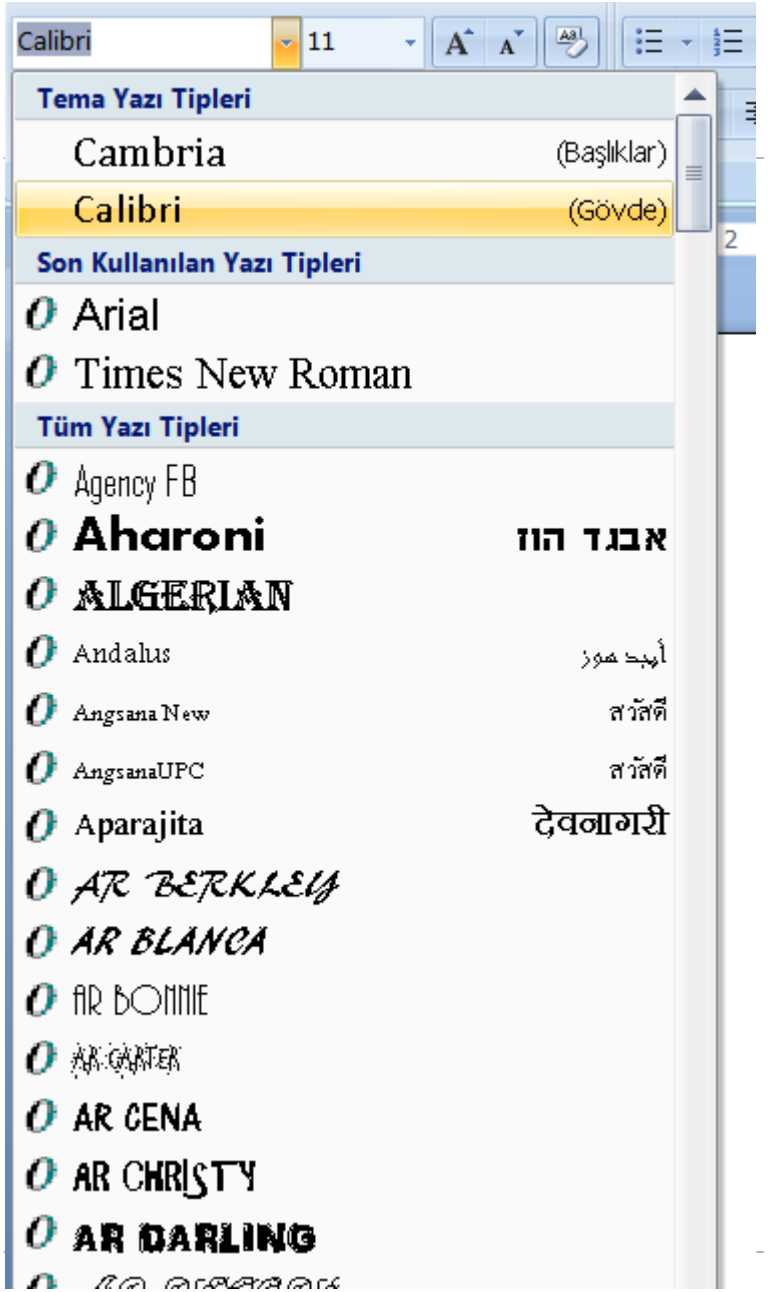
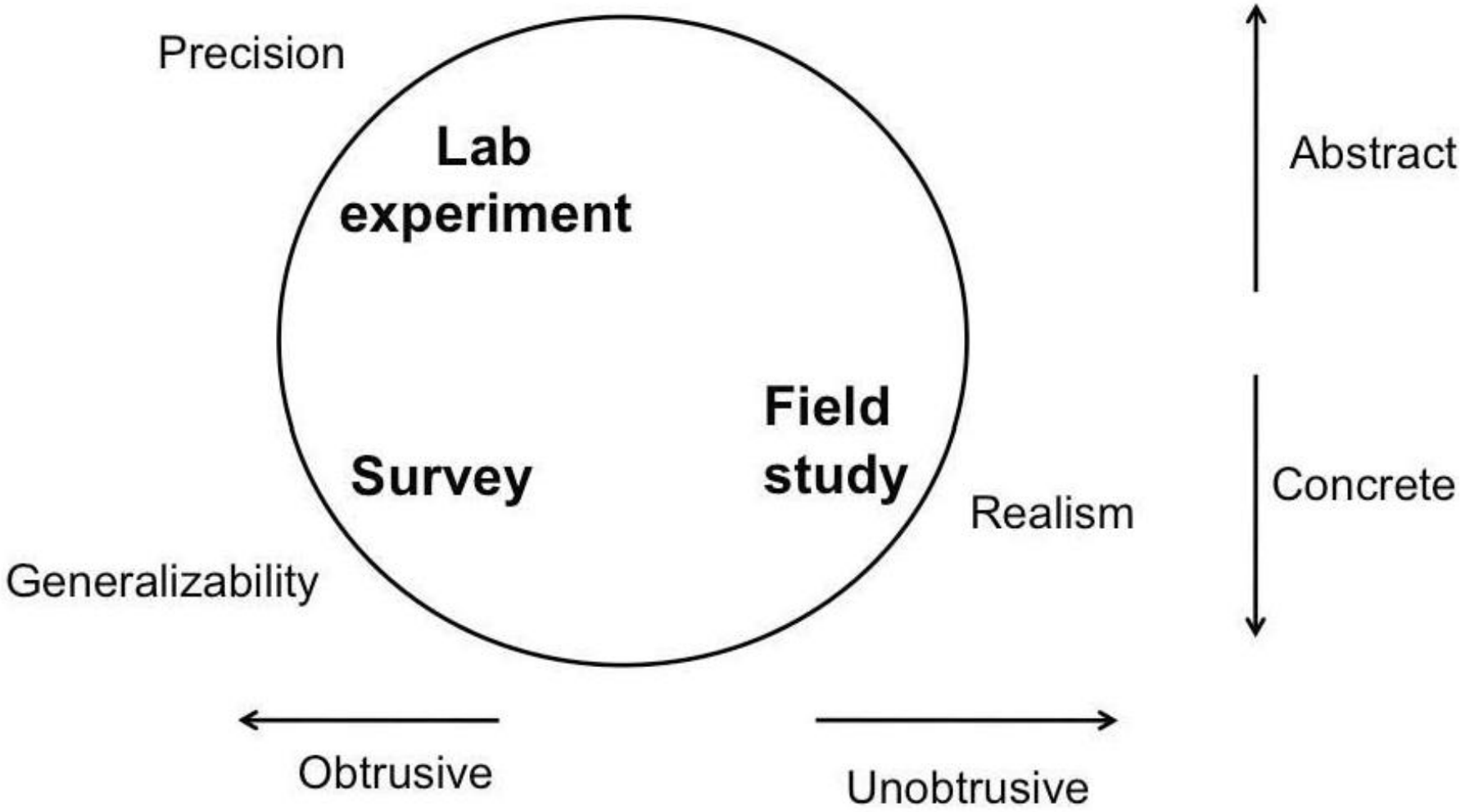


Today's Topics

- HCI research methods
- Experiment design
- Threats to validity
- Randomization, blocking, counterbalancing

Research Methods in HCI

- Lab experiment
- Field study
- Survey


Precision
Lab experiment
Field study
Survey
Realism
Generalizability
Abstract
Concrete
Obtrusive
Unobtrusive

Quantifying Usability

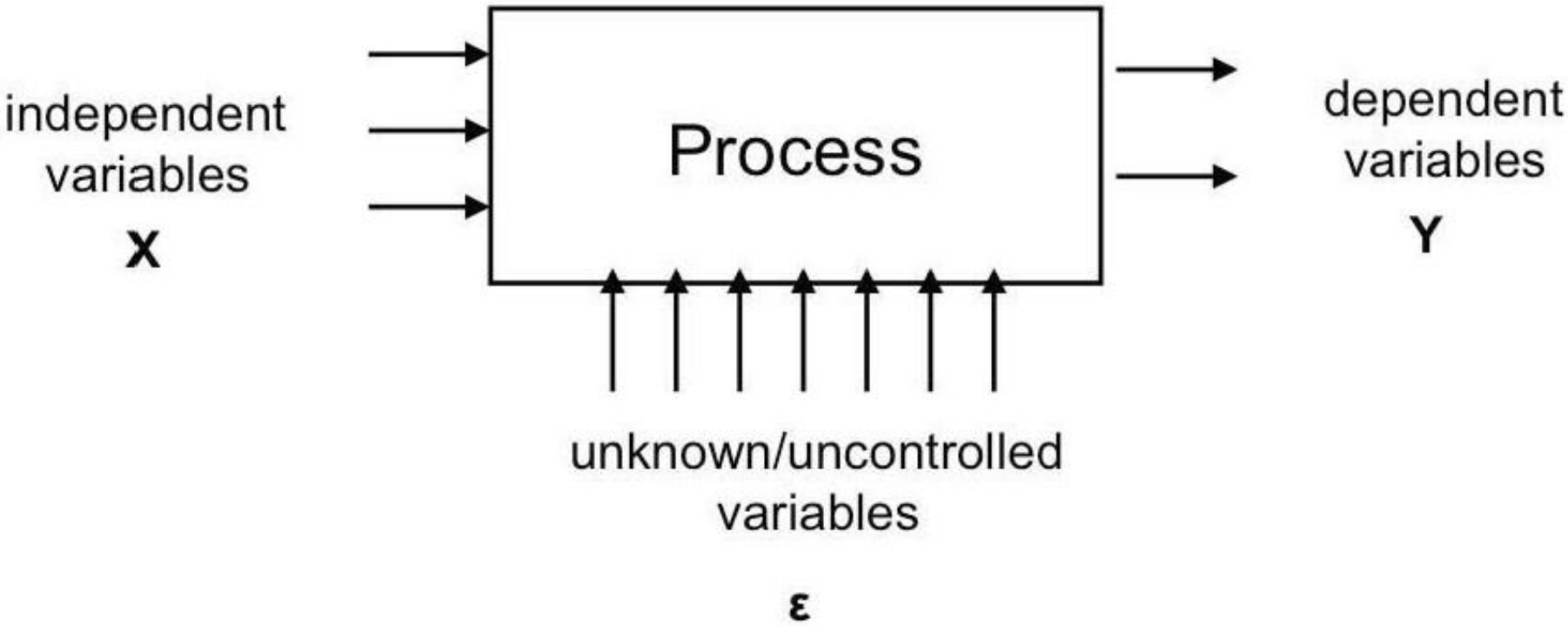
- Usability: how well users can use the system's functionality
- Dimensions of usability
  - Learnability: is it easy to learn?
  - Efficiency: once learned, is it fast to use?
  - Errors: are errors few and recoverable?
  - Satisfaction: is it enjoyable to use?

Controlled Experiment

- Start with a testable **hypothesis**
  - e.g. Mac menu bar is faster than Windows menu bar
- Manipulate **independent variables**
  - different interfaces, user classes, tasks
  - in this case, y-position of menubar
- Measure **dependent variables**
  - times, errors, # tasks done, satisfaction
- Use statistical tests to accept or reject the hypothesis

Schematic View of Experiment Design

$$Y = f(X) + \varepsilon$$



Design of the Menubar Experiment

- Users
  - Windows users or Mac users?
  - Age, handedness?
  - How to sample them?
- Implementation
  - Real Windows vs. real Mac
  - Artificial window manager that lets us control menu bar position
- Tasks
  - Realistic: word processing, email, web browsing
  - Artificial: repeatedly pointing at fake menu bar
- Measurement
  - When does movement start and end?
- Ordering
  - of tasks and interface conditions
- Hardware
  - mouse, trackball, touchpad, joystick?
  - PC or Mac? which particular machine?

Concerns Driving Experiment Design

- Internal validity
  - Are observed results actually **caused** by the independent variables?
- External validity
  - Can observed results be **generalized** to the world outside the lab?
- Reliability
  - Will consistent results be obtained by **repeating** the experiment?

Threats to Internal Validity

- Ordering effects
  - People learn, and people get tired
  - Don't present tasks or interfaces in same order for all users
  - Randomize or counterbalance the ordering
- Selection effects
  - Don't use pre-existing groups (unless group is an independent variable)
  - Randomly assign users to independent variables
- Experimenter bias
  - Experimenter may be enthusiastic about interface X but not Y
  - Give training and briefings on paper, not in person
  - Provide equivalent training for every interface
  - Double-blind experiments prevent both subject and experimenter from knowing if it's condition X or Y
    - Essential if measurement of dependent variables requires judgement

Threats to External Validity

- Population
  - Draw a random sample from your real target population
- Ecological
  - Make lab conditions as realistic as possible in important respects
- Training
  - Training should mimic how real interface would be encountered and learned
- Task
  - Base your tasks on task analysis

Threats to Reliability

- Uncontrolled variation
  - Previous experience
    - Novices and experts: separate into different classes, or use only one class
  - User differences
    - Fastest users are **10 times** faster than slowest users
  - Task design
    - Do tasks measure what you're trying to measure?
  - Measurement error
    - Time on task may include coughing, scratching, distractions
- Solutions
  - Eliminate uncontrolled variation
    - Select users for certain experience (or lack thereof)
    - Give all users the same training
    - Measure dependent variables precisely
  - Repetition
    - Many users, many trials
    - Standard deviation of the mean shrinks like the square root of N (i.e., quadrupling users makes the mean twice as accurate)

Counterbalancing

- Defeats ordering effects by varying order of conditions systematically (not randomly)
- Latin Square designs
  - randomly assign subjects to equal-size groups
  - A,B,C,... are the experimental conditions
  - Latin Square ensures that each condition occurs in every position in the ordering for an equal number of users

G1	G2
A	B
B	A

2x2

G1	G2	G3
A	C	B
B	A	C
C	B	A

3x3

G1	G2	G3	G4
A	D	C	B
B	A	D	C
C	B	A	D
D	C	B	A

4x4

... etc.

Summary

- Research methods in HCI include lab experiments, field studies, and surveys
- Controlled experiments manipulate independent variables and measure dependent variables
- Must consider and defend against threats to internal validity, external validity, and reliability
- Blocking, randomization, and counterbalancing techniques help